

EduSense Week 2 and 3 Resources

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May 2026

1 Introduction

Welcome to Week 2 of the EduSense project!

In Week 1, the primary focus was on understanding the project idea, exploring the overall pipeline, and getting familiar with the tools that will be used throughout the project. From this week onwards, we will start diving into the technical components that form the foundation of our AI-powered learning assistant.

The goal of EduSense is to build a system that can listen to classroom lectures, convert speech into text, generate meaningful summaries, and eventually assist students in reviewing and understanding lecture content efficiently. To achieve this, we need to understand both machine learning fundamentals and the tools commonly used in modern AI applications.

For the next 2–3 weeks, we will focus on two important topics:

- **PyTorch** – A widely used deep learning framework that provides the building blocks required to train and deploy machine learning models.
- **OpenAI Whisper** – A state-of-the-art automatic speech recognition (ASR) model capable of converting audio recordings into text with high accuracy.

By the end of this phase, you should be comfortable working with neural network frameworks, understanding model pipelines, and using pre-trained models for speech-to-text tasks.

Please go through the provided resources carefully and try to experiment with the code on your own. Hands-on practice is often the fastest way to learn.

Keep checking the GitHub repository regularly for assignments, updates, and additional resources.

2 PyTorch

PyTorch is an open-source machine learning framework developed primarily for deep learning research and applications. It provides an intuitive interface for

creating neural networks, training models, and performing tensor computations efficiently on CPUs and GPUs.

For this project, PyTorch will serve as the backbone for most machine learning-related tasks. Understanding its basic workflow will be useful not only for EduSense but also for future AI and ML projects.

Resources

- Tutorial link
Pytorch tutorial

Please note to look at only the relevant sections.

3 OpenAI Whisper

Whisper is a large-scale automatic speech recognition (ASR) model developed by OpenAI and trained on hundreds of thousands of hours of multilingual and multitask audio data. It is capable of performing speech-to-text transcription, language identification, and translation across multiple languages with high robustness to accents, background noise, and varying recording conditions.

In the EduSense pipeline, Whisper serves as the front-end speech processing module. The lecture audio is provided as input to the model, which generates a textual transcript that can subsequently be processed by downstream natural language processing components such as summarization, question answering, and note generation systems.

Whisper is based on the Transformer architecture, one of the most influential developments in modern machine learning. Unlike traditional recurrent models, Transformers rely on the **self-attention mechanism**, which enables the model to capture long-range dependencies and contextual relationships within sequential data. In the case of Whisper, the input audio signal is first converted into a log-Mel spectrogram representation, which is then processed by a Transformer encoder. A Transformer decoder subsequently generates the corresponding text sequence token by token.

Rather than training an ASR model from scratch, which requires massive computational resources and large annotated speech datasets, we will leverage Whisper's pre-trained capabilities and integrate it directly into our system. This allows us to focus on building and optimizing the higher-level components of the EduSense pipeline while utilizing a state-of-the-art speech recognition model.

Resources

- GitHub Repository:
OpenAI Whisper Repository

- Documentation / Tutorial:
Whisper Documentation

If you want to know more about how the architecture works or the math behind it, we encourage you to search in youtube. You will find channels with wonderful explanations like 3Blue1Brown. The paper about this mechanism, "Attention is all you need" gained wide popularity in the field of machine learning.

4 Expected Outcomes

By the end of this 2 week's learning phase, you should be able to:

- Understand the basics of PyTorch tensors and neural networks.
- Load and run pre-trained models.
- Use Whisper to transcribe audio files.
- Understand where speech-to-text fits into the EduSense pipeline.
- Begin exploring how different AI components can be connected to form a complete application.